

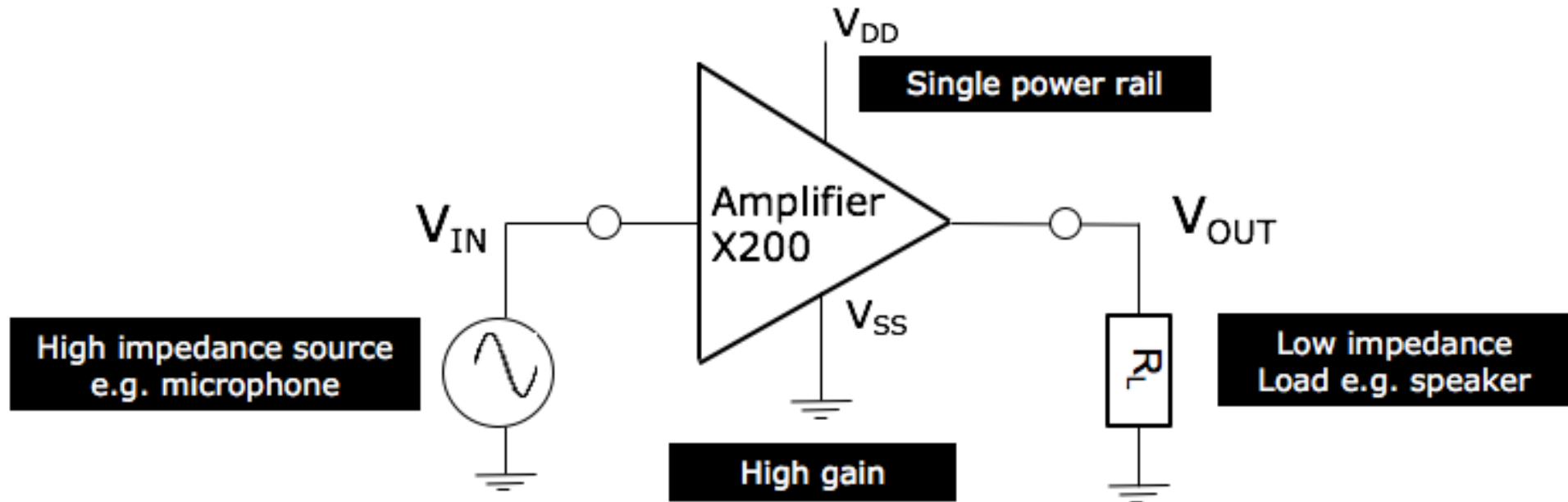
# Lecture 2

## Amplification and Single-rail Op-amp

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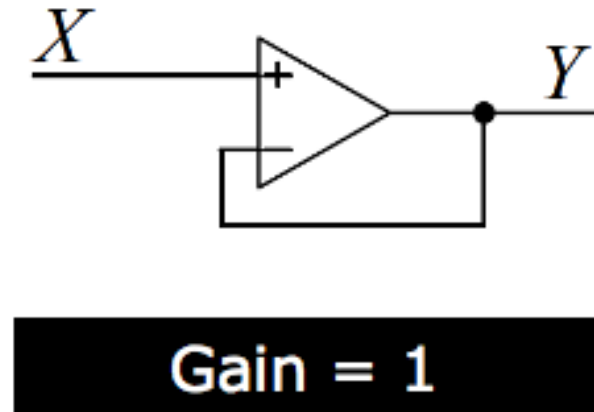
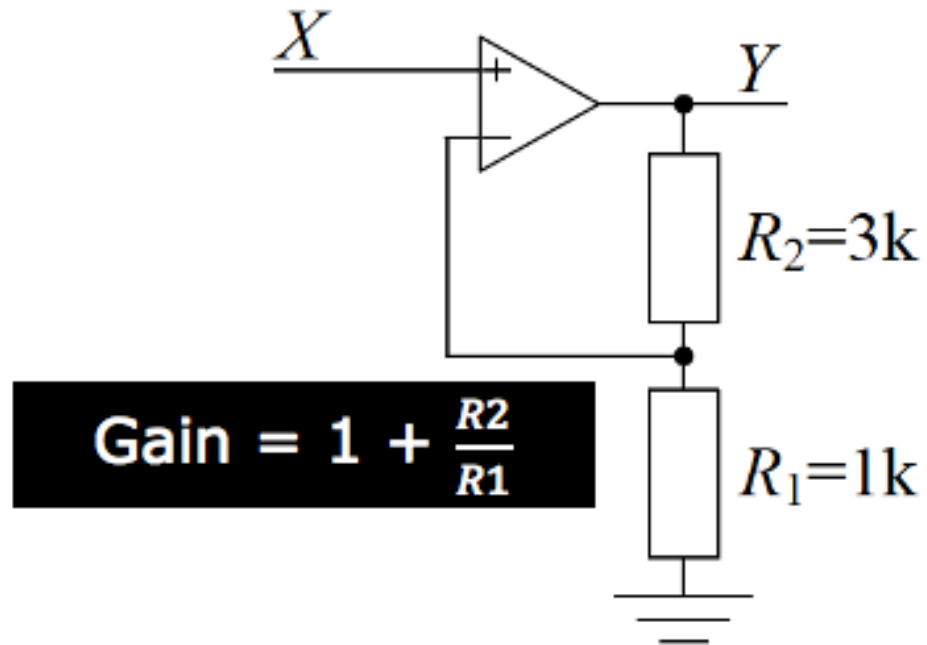
URL: <https://elec50001-cas.github.io/course-website/>  
E-mail: [p.cheung@imperial.ac.uk](mailto:p.cheung@imperial.ac.uk)

# What you will learn in Lectures 2 and 3?



- ❖ Small input signal
- ❖ High gain amplification
- ❖ Working with single power supply rail
- ❖ Low impedance load (next Lecture)

# Non-inverting Amplifier & Voltage Follower



- ❖ Year 1 circuits module, part 1, lecture 9, slides 11 and 12
- ❖ Non-inverting amplifier using op-amp, Gain x4
- ❖ Special case:  $R_2 = 0$ ,  $R_1 = \infty$ , Gain = 1
  - Voltage follower or unity gain buffer

# Real-life Op Amp



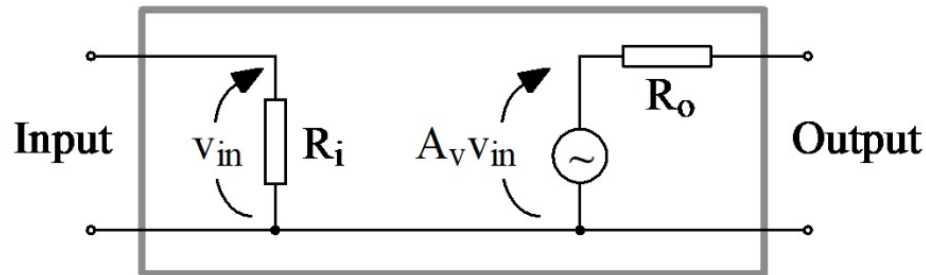
1 MHz, Low-Power Op Amp

## Description

The Microchip Technology Inc. MCP6001/2/4 family of operational amplifiers (op amps) is specifically designed for general-purpose applications. This family has a 1 MHz Gain Bandwidth Product (GBWP) and 90° phase margin (typical). It also maintains 45° phase margin (typical) with a 500 pF capacitive load. This family operates from a single supply voltage as low as 1.8V, while drawing 100  $\mu$ A (typical) quiescent current. Additionally, the MCP6001/2/4 supports rail-to-rail input and output swing, with a common mode input voltage range of  $V_{DD} + 300$  mV to  $V_{SS} - 300$  mV. This family of op amps is designed with Microchip's advanced CMOS process.

- ❖ Limited to 1MHz signal frequency (GBP) (not infinite gain at all frequencies)
- ❖ Stable under high capacitance load (linked to phase margin)
- ❖ **Single power supply operation**
- ❖ **Rail-to-rail input/output swing**
- ❖ Low supply current when idle
- ❖ Near rail-to-rail common mode input voltage

# MCP6001/2/4 as a near ideal op-amp



✓  $Z_{in}$  of MCP6001 is very large

|                              |            |                |                |
|------------------------------|------------|----------------|----------------|
| Common Mode Input Impedance  | $Z_{CM}$   | $10^{13}    6$ | $\Omega    pF$ |
| Differential Input Impedance | $Z_{DIFF}$ | $10^{13}    3$ | $\Omega    pF$ |

## ❖ Ideal op-amp

- $R_i \rightarrow \infty$
- $A_v \rightarrow \infty$
- $R_o \rightarrow 0$

✓ Input current is negligible

|                        |       |           |    |                      |
|------------------------|-------|-----------|----|----------------------|
| Input Bias Current:    | $I_B$ | $\pm 1.0$ | pA |                      |
| Industrial Temperature | $I_B$ | 19        | pA | $T_A = +85^\circ C$  |
| Extended Temperature   | $I_B$ | 1100      | pA | $T_A = +125^\circ C$ |

✓ Gain is near infinite (true at low frequency)

|                                  |          |                          |    |
|----------------------------------|----------|--------------------------|----|
| DC Open-Loop Gain (Large Signal) | $A_{OL}$ | 112<br>$> 1 \times 10^5$ | dB |
|----------------------------------|----------|--------------------------|----|

✓ Output impedance is low (specified as  $I_{SC}$ )

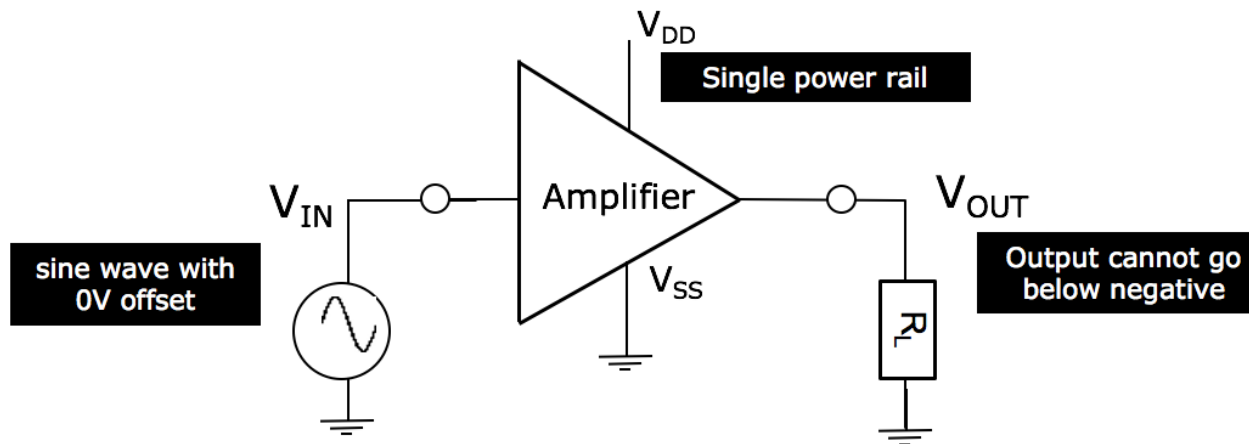
|                              |          |                     |          |                                    |
|------------------------------|----------|---------------------|----------|------------------------------------|
| Output Short Circuit Current | $I_{SC}$ | $\pm 6$<br>$\pm 23$ | mA<br>mA | $V_{DD} = 1.8V$<br>$V_{DD} = 5.5V$ |
|------------------------------|----------|---------------------|----------|------------------------------------|

# Problem with single supply rail

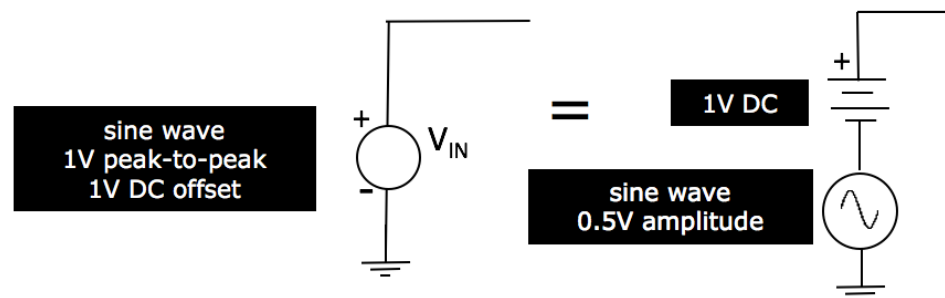
- ❖ Output swing limited to rail voltages

|                              |                  |                                |    |
|------------------------------|------------------|--------------------------------|----|
| Maximum Output Voltage Swing | $V_{OL}, V_{OH}$ | $V_{SS} + 25$<br>$V_{DD} - 25$ | mV |
|------------------------------|------------------|--------------------------------|----|

- ❖ Input cannot be a sinewave with 0V DC offset

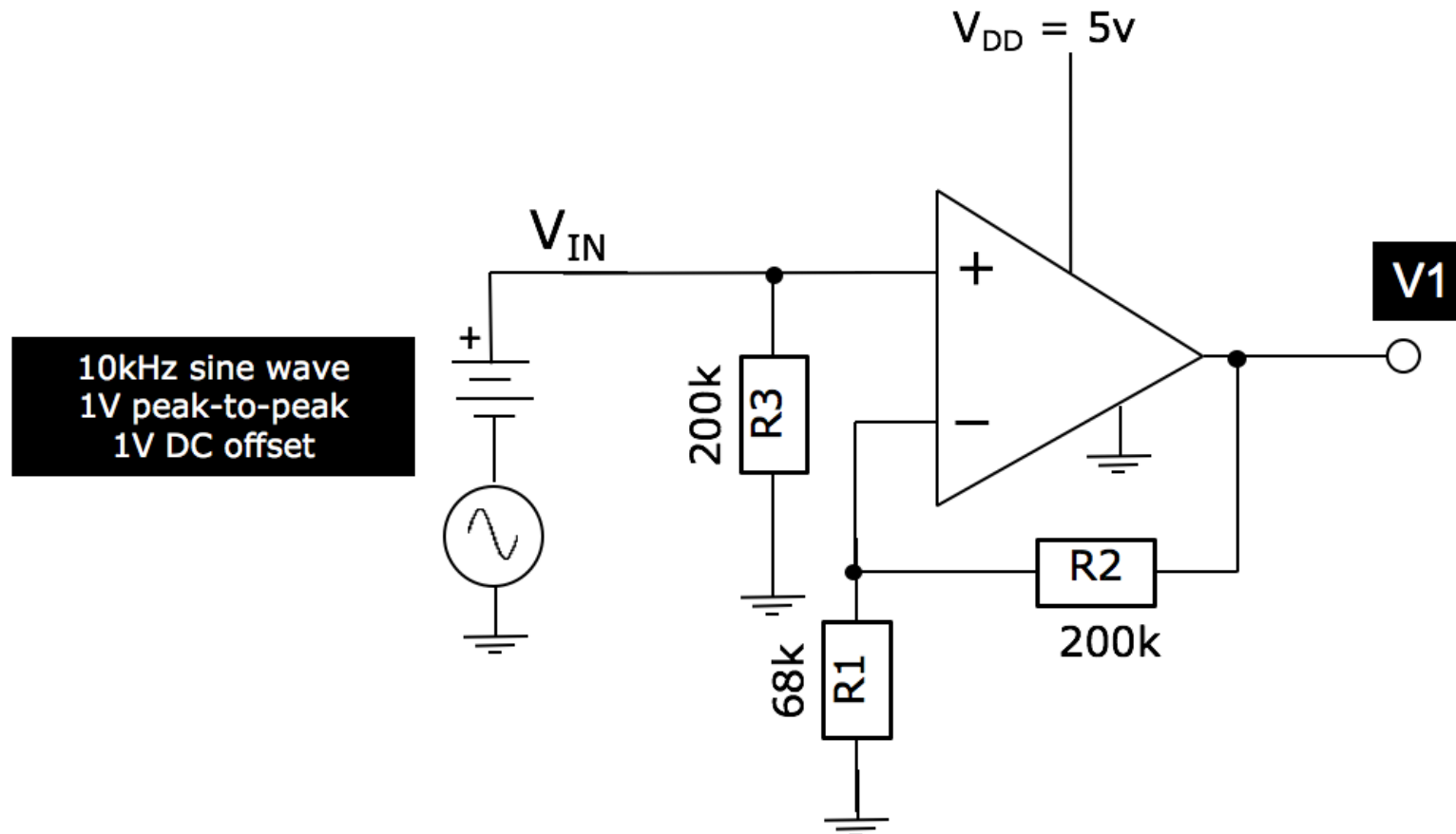


- ❖ Need to add DC offset to input



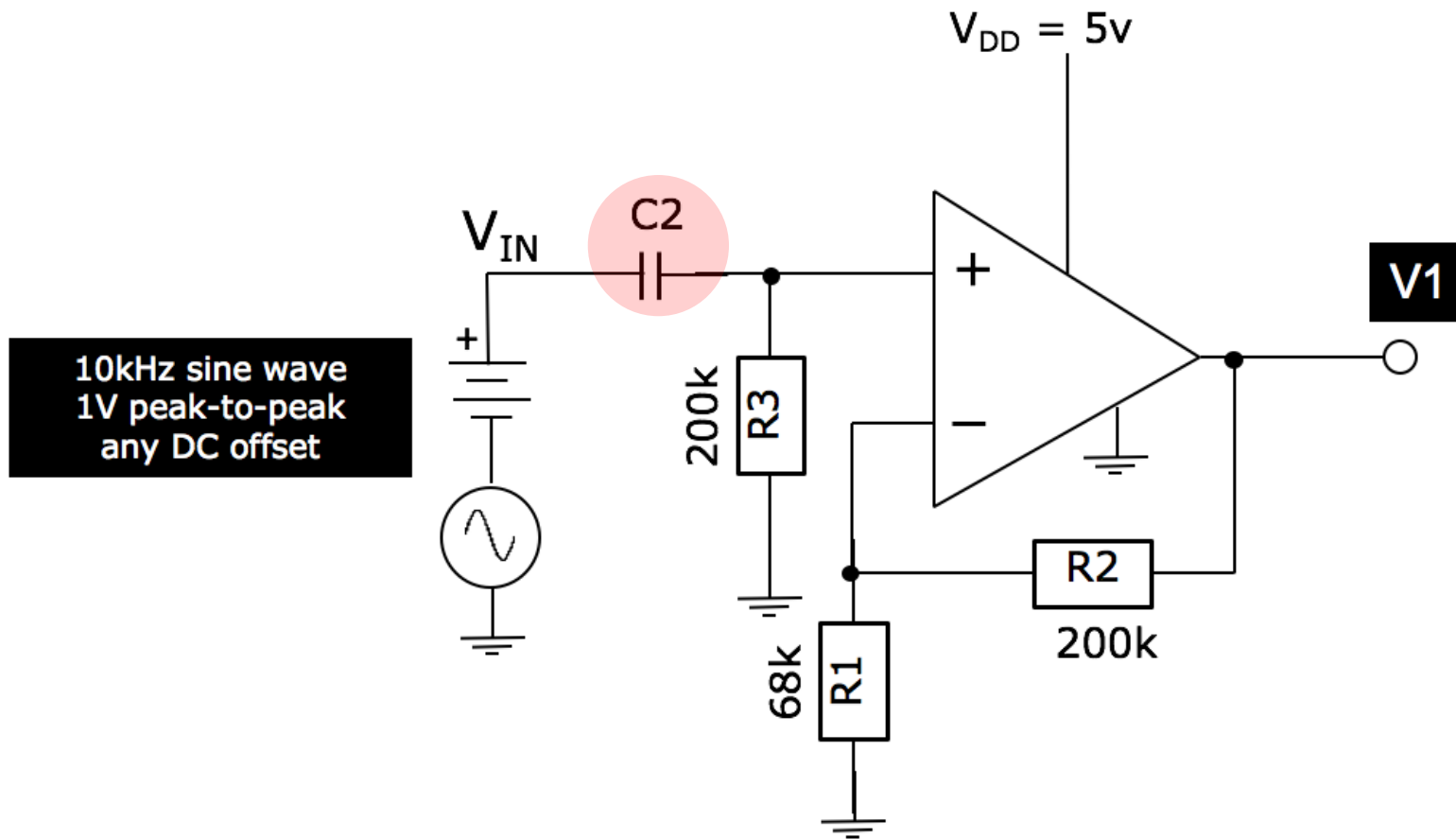
## X4 amplifier fail

- ❖ Amplifier Gain = +4, output swing = 2V to 6V – not possible



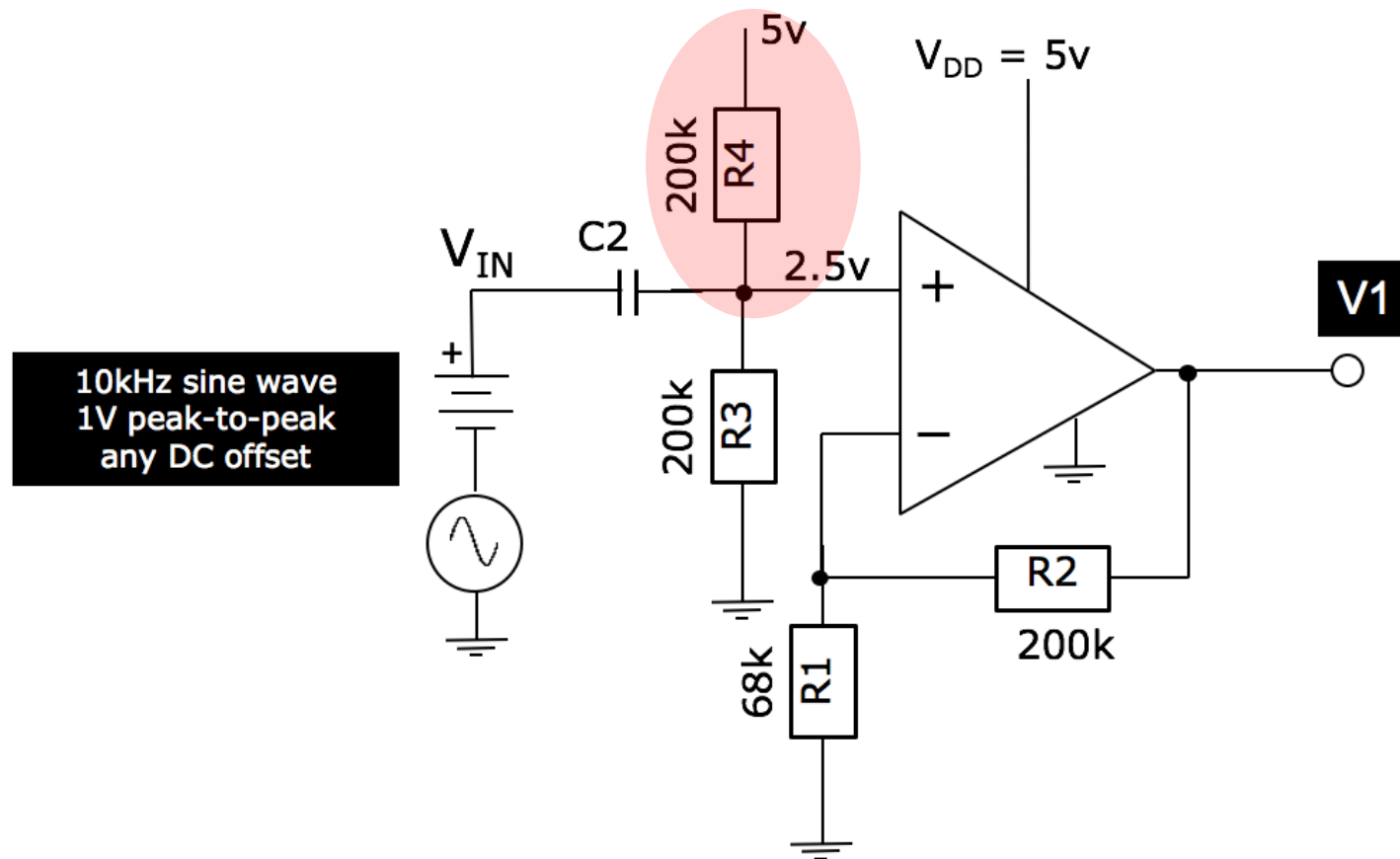
# Attempt 1 – AC coupling signal

- ❖ AC couple input signal – block DC offset to amplifier
- ❖ Does not work for single supply op-amp



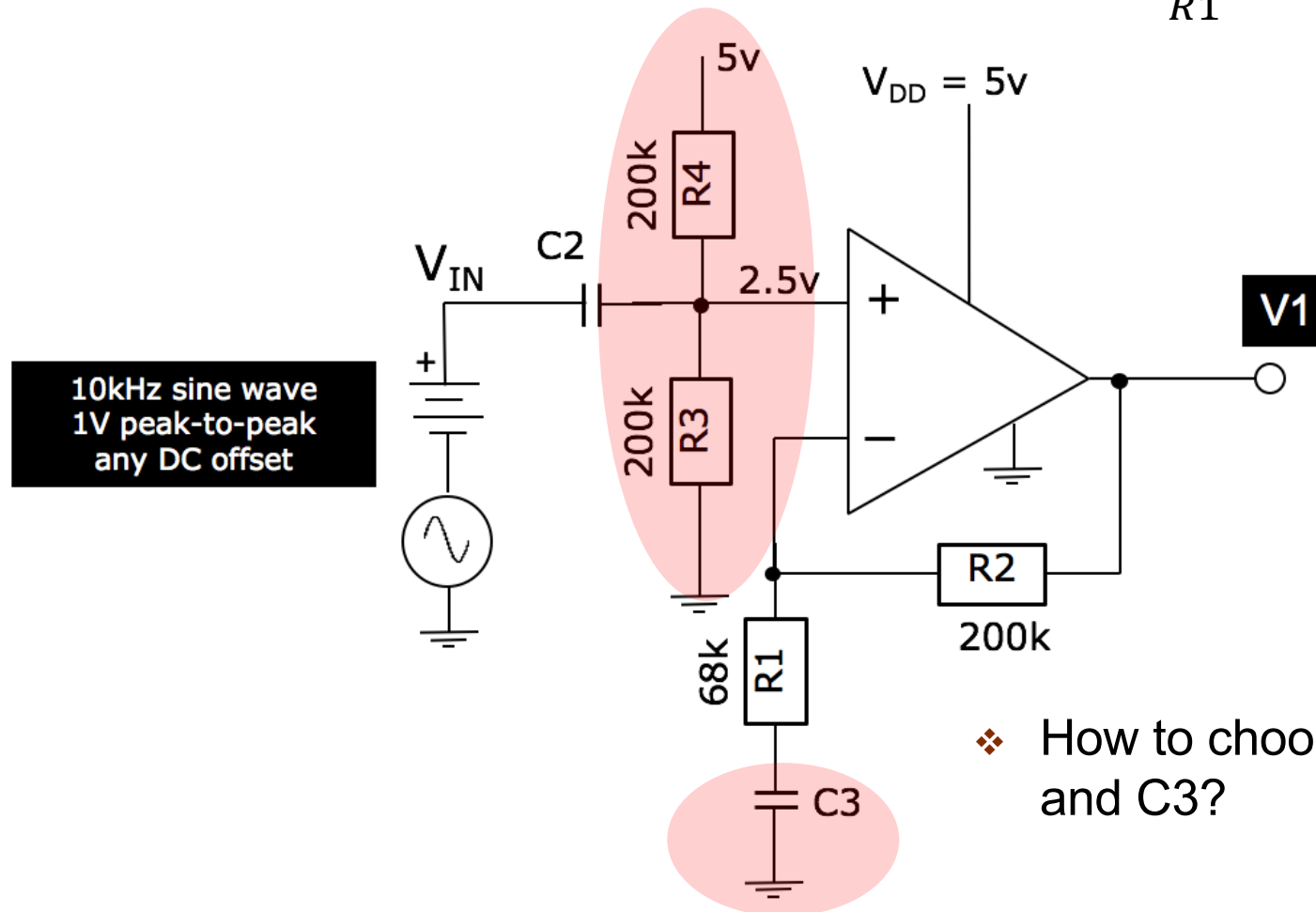
## Attempt 2 – Add bias to V+ input

- ❖ Bias input V+ to half of power supply rail voltage
  - Maximize input voltage swing
  - Still not working – bias voltage is now the unwanted DC offset!



# Final working solution

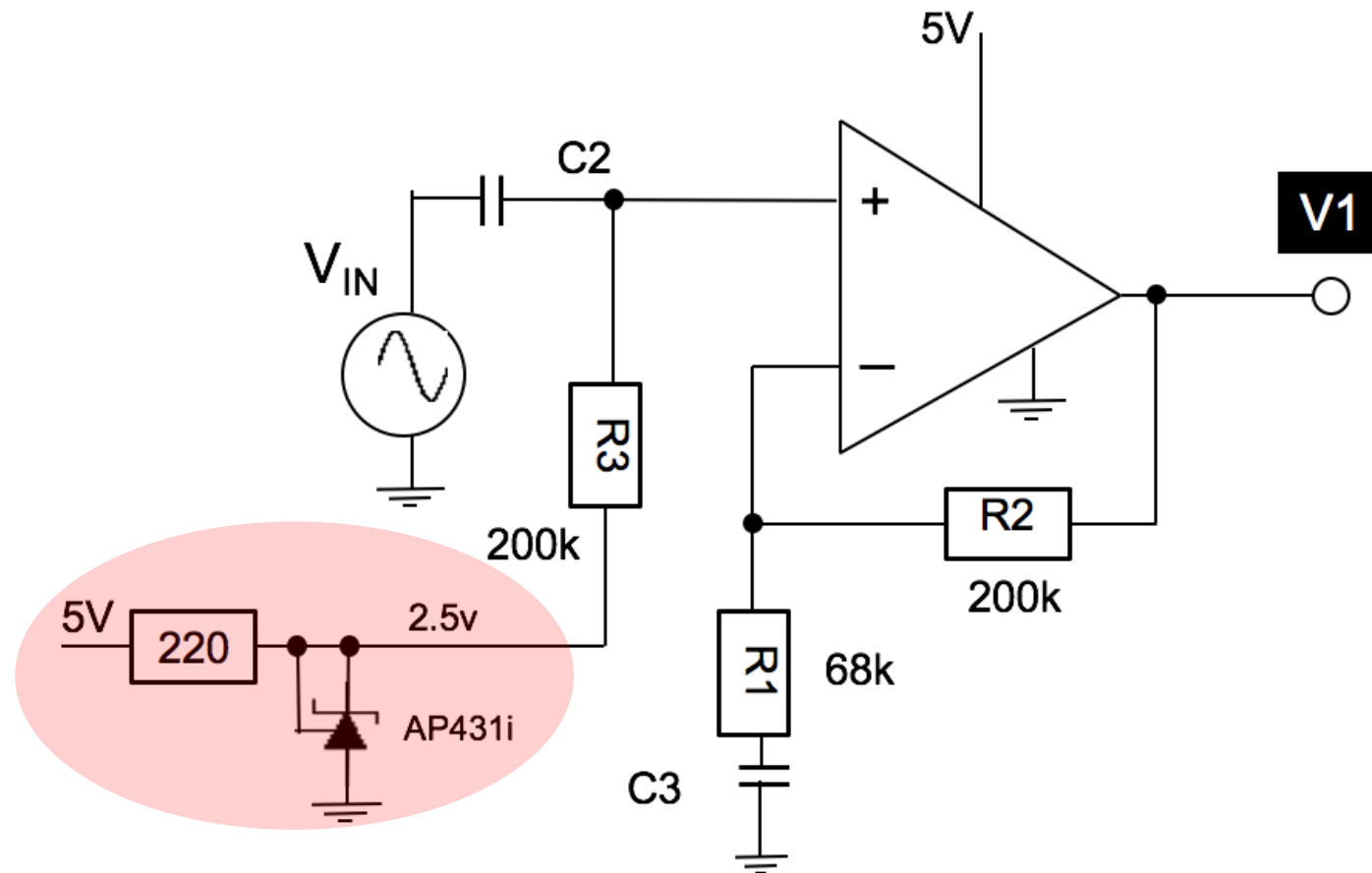
- ❖ Final solution: Add C3 such that gain of op-amp = 1 at 0 Hz (DC)
- ❖ Gain of op-amp at valid signal frequency is  $G = 1 + \frac{R2}{R1}$



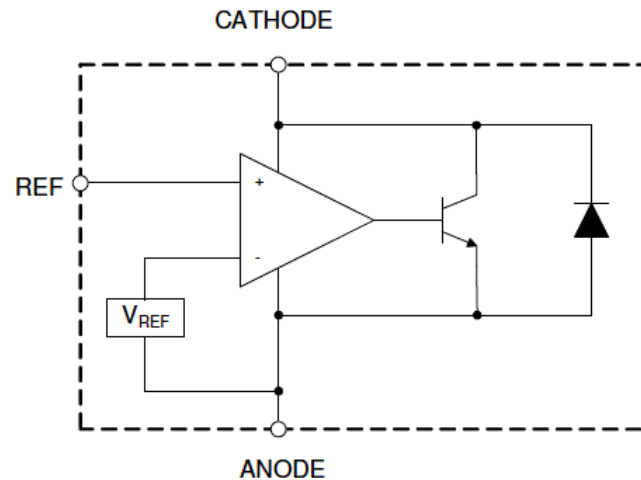
- ❖ How to choose values for C2 and C3?

# Better Bias Circuit

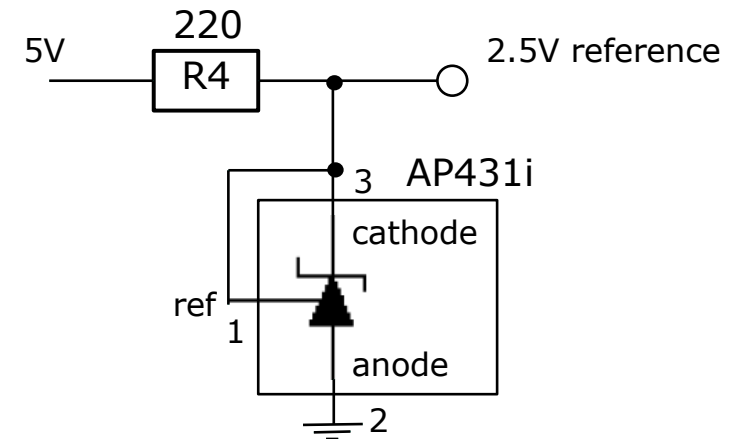
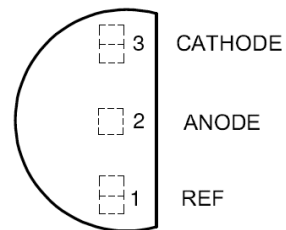
- ❖ Generate bias voltage using a voltage reference circuit, e.g. AP431i
- ❖ Bias voltage NOT susceptible to noise on 5V supply



# AP431i voltage reference in detail



AP431i  
(Top View)

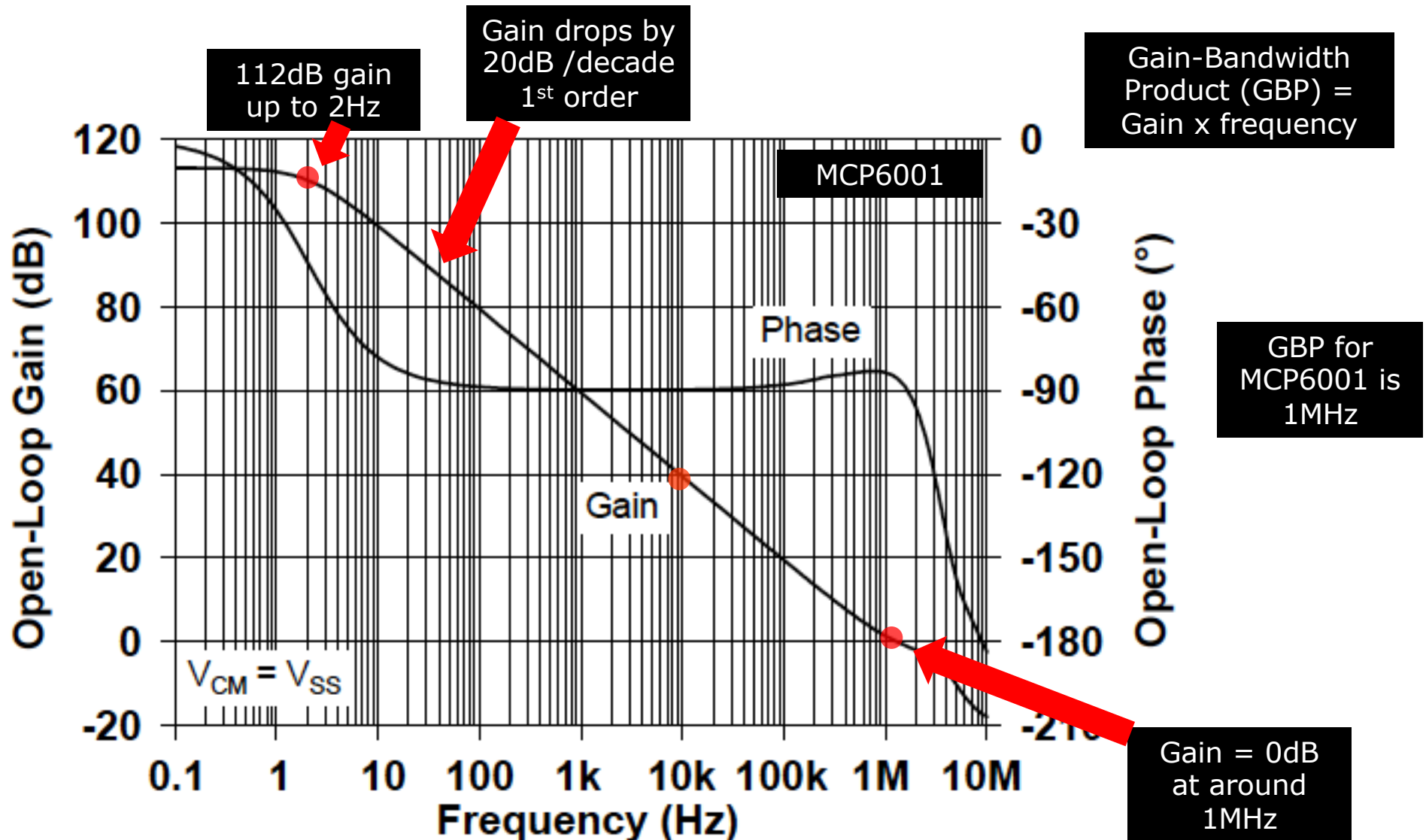


|          |                 | Min | Max |    |
|----------|-----------------|-----|-----|----|
| $I_{KA}$ | Cathode Current | 0.1 | 100 | mA |

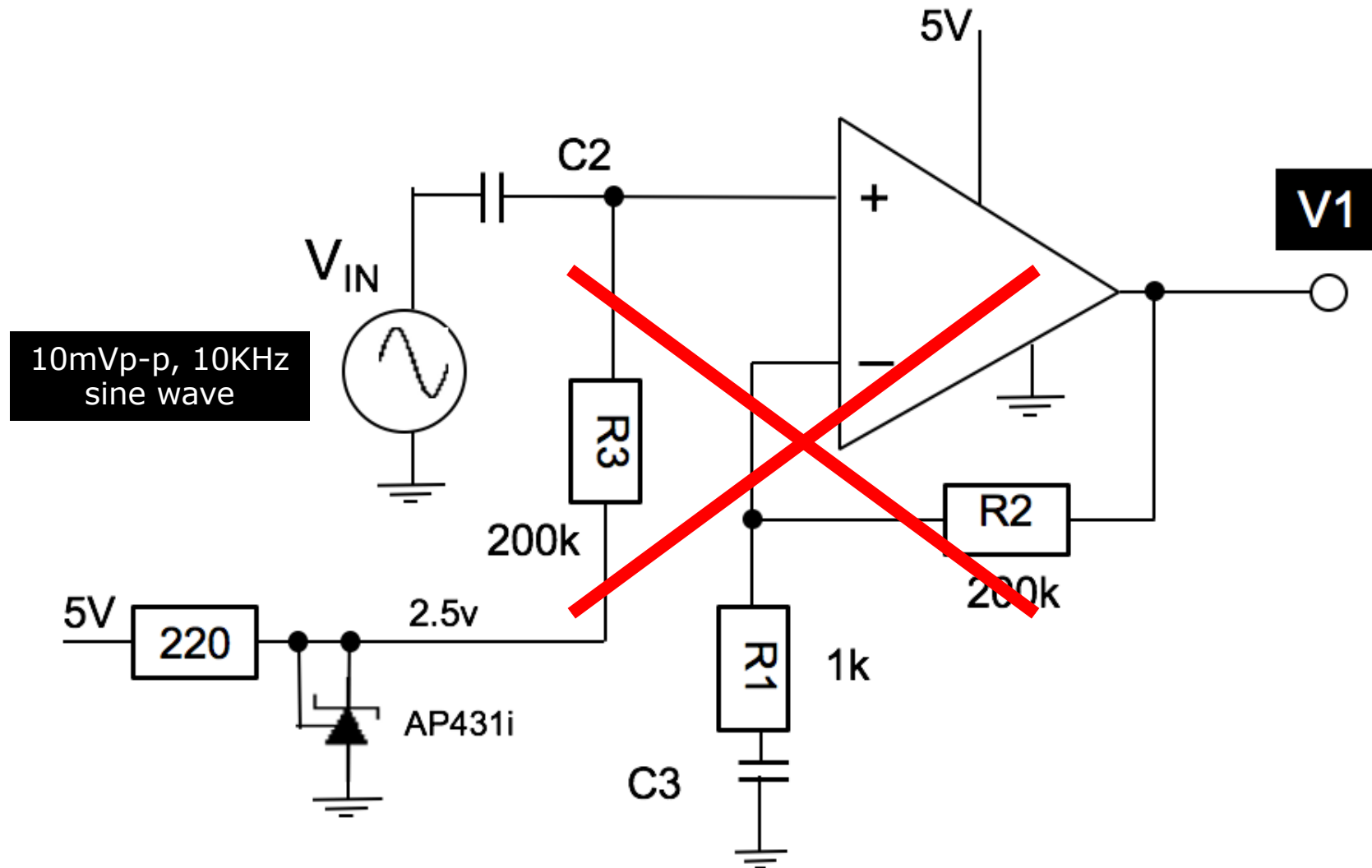
|                 |                   |                                  | Min   | Typical | Max   |   |
|-----------------|-------------------|----------------------------------|-------|---------|-------|---|
| $V_{REF} \ 1\%$ | Reference Voltage | $V_{KA} = V_{REF}, I_{KA} = 1mA$ | 2.475 | 2.500   | 2.525 | V |

|          |                   |                                                                      | Typical | Max |          |
|----------|-------------------|----------------------------------------------------------------------|---------|-----|----------|
| $Z_{KA}$ | Dynamic Impedance | $V_{KA} = V_{REF},$<br>$I_{KA} = 1 \text{ to } 100mA, f \leq 1.0kHz$ | 0.1     | 0.3 | $\Omega$ |

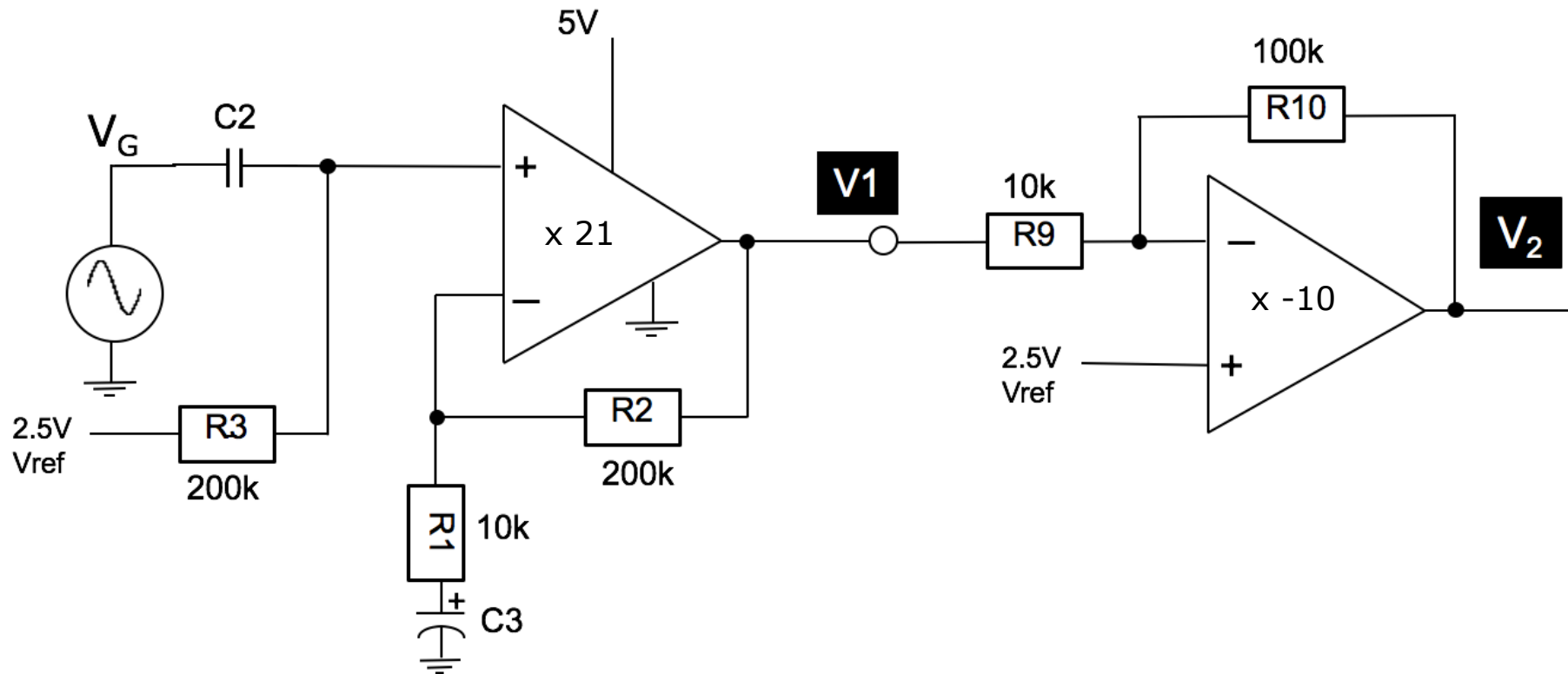
# Open-loop Gain vs Frequency for MCP6001



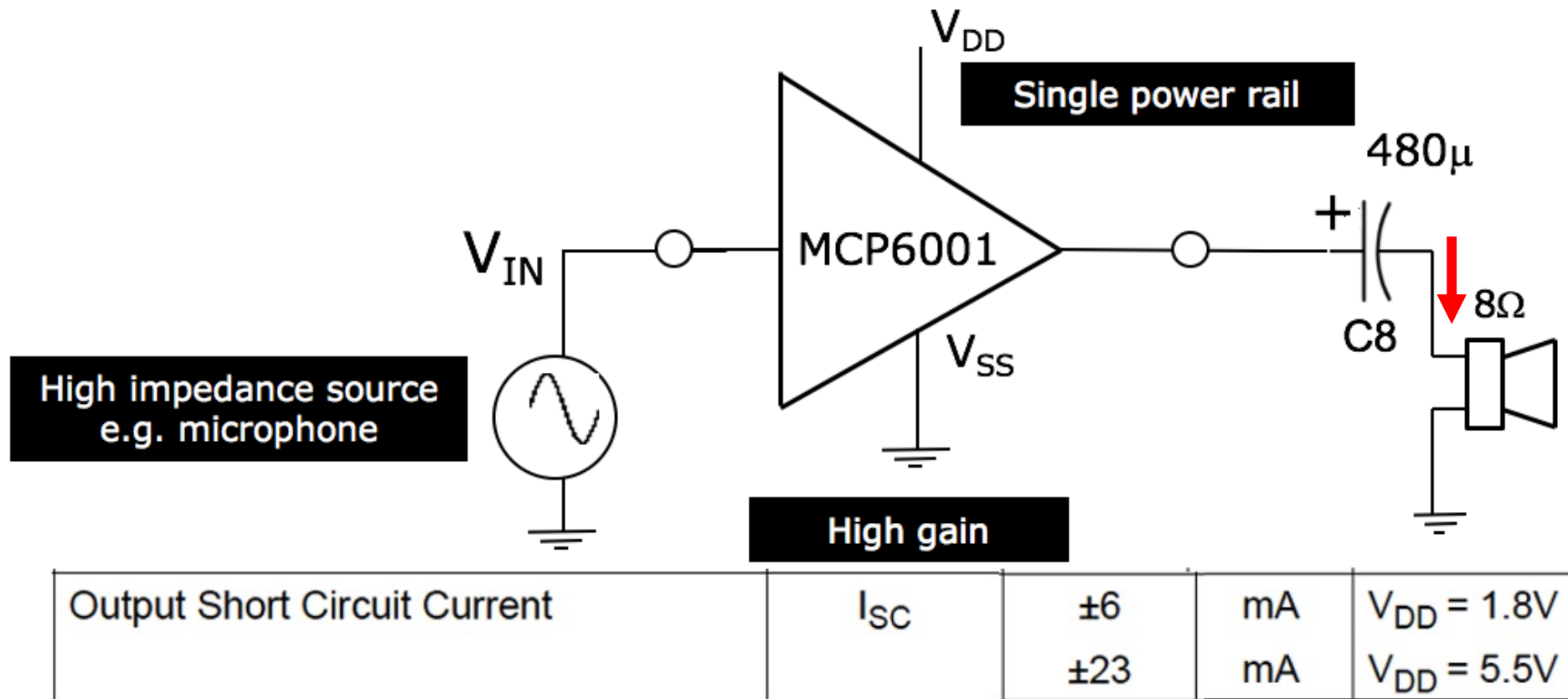
# Amplify 10kHz signal with Gain of 200



# Solution - Two stage amplification



# Problem with driving low impedance load



- ❖ Needs AC coupling due to output bias voltage of 2.5V
- ❖ Maximum current is over 600mA!
- ❖ MCP6001 cannot drive 8 $\Omega$  speaker